

shine® **Joinflex Forte Film Coated Tablet**

Active Ingredient(s):

Each film coated tablet contains:

Glucosamine Sulphate 2KCl	500mg
(eq. to Glucosamine Sulphate 377mg)	
Chondroitin Sulphate	400mg

Preservative(s):

Methyl Paraben	0.023%
Propyl Paraben	0.005%
Ethyl Paraben	0.005%

Pharmacology (Summary of Pharmacokinetics and Pharmacodynamics):

Pharmacodynamics:

Glucosamine sulphate seems to be capable of interfering with various pathological mechanisms of osteoarthritis, stimulating synthetic activity of the chondrocytes and inhibiting the chondrolysis provoked by the degrading enzyme released into the articular cavity during osteoarthritis. Glucosamine has the ability to actually provide the basic raw material necessary to maintain a healthy cartilage and also to prevent degeneration of the cartilage, which especially occurs in conditions like osteoarthritis.

The benefit of chondroitin sulphate in patients with osteoarthritis is likely the result of a number of effects including its anti-inflammatory activity, the stimulation of the synthesis of proteoglycans and hyaluronic acid, and the decrease in catabolic activity of chondrocytes inhibiting the synthesis of proteolytic enzymes, nitric oxide and other substances that contribute to damage cartilage matrix and cause death of articular chondrocytes. A recent review summarizes data from relevant reports describing the biochemical basis of the effect of chondroitin sulphate on osteoarthritis articular tissues. The rationale behind the use of chondroitin sulphate is based on the belief that osteoarthritis is associated with a local deficiency in some natural substances, including chondroitin sulphate.

Recently, new mechanisms of action have been described for chondroitin sulphate. In an in vitro study, chondroitin sulphate reduced the IL-1 β -induced nuclear factor-kB (NF-kB) translocation in chondrocytes. In addition, chondroitin sulphate has recently shown a positive effect on osteoarthritic structural changes occurred in the subchondral bone.

Pharmacokinetics:

After oral administration of glucosamine sulphate, the total bioavailability following first pass metabolism was found to be 26%. Intramuscular bioavailability was noted to be approximately 96%. After the initial period of absorption, the drug is distributed by the blood stream and the major sites of distribution were the liver, kidneys and articular cartilage.

Pharmacokinetic studies performed on humans and experimental animals after oral administration of chondroitin sulphate revealed that it can be absorbed orally. Chondroitin sulphate shows first-order kinetics up to single doses of 3,000 mg. Multiple doses of 800 mg in patients with osteoarthritis do not alter the kinetics of chondroitin sulphate. The bioavailability of chondroitin sulphate ranges from 15% to 24% of the orally administered dose. More particularly, on the articular tissue, it is reported that chondroitin sulphate is not rapidly absorbed in the gastro-intestinal tract and a high content of labeled chondroitin sulphate is found in the synovial fluid and cartilage.

Indication(s):

As adjuvant therapy for osteoarthritis.

Dosage and Administration:

Adults: Take 4 tablets daily, in single dose or in divided doses.

Route of Administration:

For oral use.

Warning and Precautions:

1. Avoid self-diagnosing and self-medicating. Persons with stiff or sore joints should consult a doctor for an evaluation, to be sure that the problem is indeed osteoarthritis.
2. High dosages may cause nausea, diarrhea, indigestion, and heartburn. It should be taken with meals to help avoid these problems.

- Those with Type 2 Diabetes Mellitus and those who are overweight and have problems with glucose tolerance should have their blood sugars carefully monitored if they use glucosamine supplements.
- Because of insufficient safety data, children, pregnant woman, and nursing mothers should avoid using glucosamine.

Interaction with Other Medicaments:

Shine Joiflex Forte Film Coated Tablet should not be taken with heart medications or insulin. Those taking diuretics may require higher dosages on a daily basis. It is suggested that because chondroitin sulphate molecule is similar to the heparin molecule, its use together with anticoagulant drugs should be avoided.

Pregnancy and Lactation:

Use of glucosamine and chondroitin should be avoided during pregnancy and lactation.

Contraindications:

Diabetics need to be aware that glucosamine is a glucose-containing food, and can raise glucose levels.

Contraindicated in patients having hypersensitivity to any of the ingredients of this formulation.

Hypersensitivity to glucosamine or chondroitin and patients who are allergic to seafood.

Side Effect(s) / Adverse Reaction(s):

- Cardiovascular: Peripheral oedema, tachycardia were reported in a few patients following larger clinical trials investigating oral administration in osteoarthritis. Causal relationship has not been established.
- Central nervous system: Drowsiness, headache, insomnia have been observed rarely during therapy (less than 1%).
- Gastrointestinal: Nausea, vomiting, diarrhea, dyspepsia, or epigastric pain, constipation, heartburn and anorexia have been described rarely during oral therapy with glucosamine.
- Skin: Skin reactions such as erythema and pruritus have been reported with therapeutic administration of glucosamine.

Symptoms and Treatment for Overdosage, and Antidote(s):

Not known.

Shelf-Life:

3 years from the date of manufacture.

Storage Condition(s):

Store at temperature below 30°C. Protect from light and moisture.

Product Description:

Dark brown opaque coloured, oblong shaped, gelatin coated tablets.

Dosage Forms and Packaging Available:

Blister packing of 15's x 2, 15's x 4, 15's x 6 and 15's x 8.

Not all pack sizes may be marketed.

Manufacturer:

GELTEC PVT. LTD.

Sr #24, 26/3, 27/2, Yadavanahalli, Attibele,
Bangalore-562107, India.

Product Registration Holder and Distributor:

Y.S.P. INDUSTRIES (M) SDN. BHD. (199001001034)

Lot 3, 5, 7, 12 & 14, Jalan P/7, Section 13,
Kawasan Perindustrian Bandar Baru Bangi,
43000 Kajang, Selangor, Malaysia.

Product Info Line: 1 800 88 3679

Ordering Line: 1 800 88 3027

